Forecasting Drought-to-Flood Transitions at Subseasonal Timescales

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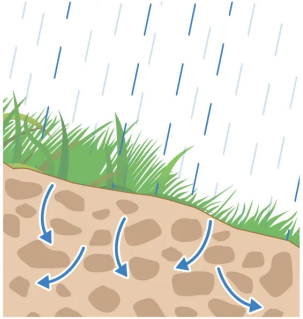
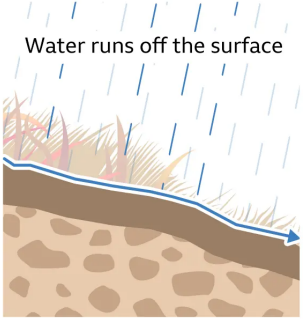
What Happens When Drought Turns Into Flood?

Normal conditions



Soil absorbs water like a sponge

Drought conditions



Hard layer of soil repels water

Adapted from BBC Research (2022)

24%

of all floods globally follow drought

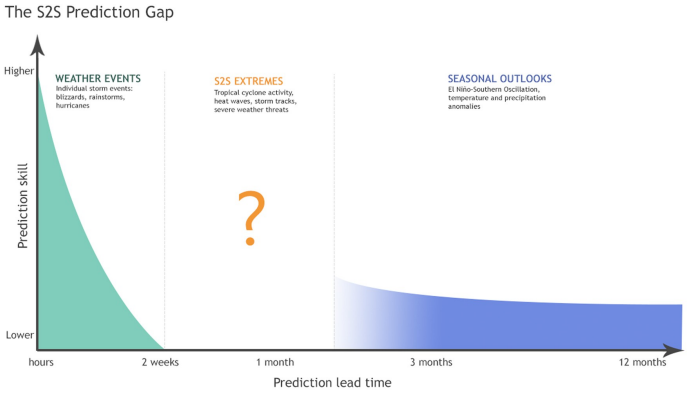
Matanó et al. (2024)

~6 weeks

average transition time fastest under a month

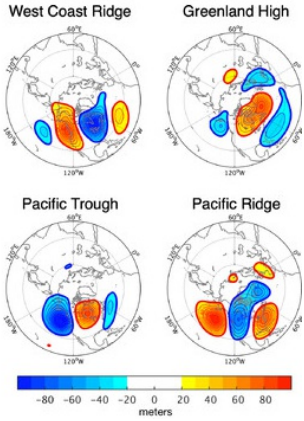
Yang et al. (2025)

Why Is This So Hard to Forecast?



Adapted from Mariotti et al. (2018)

Weather regimes



Adapted from DeFlorio et al. (2024)

What Causes These Transitions?

Persistent blocking

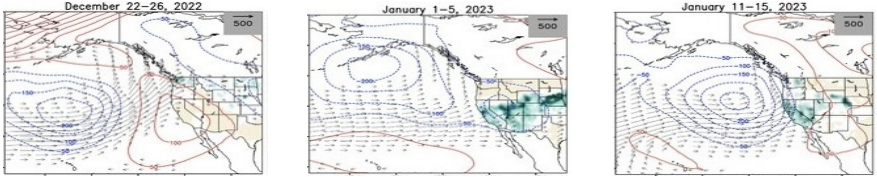
Rainfall suppressed
Soil dries out

Blocking breaks down

Flow shifts
Moist air moves in

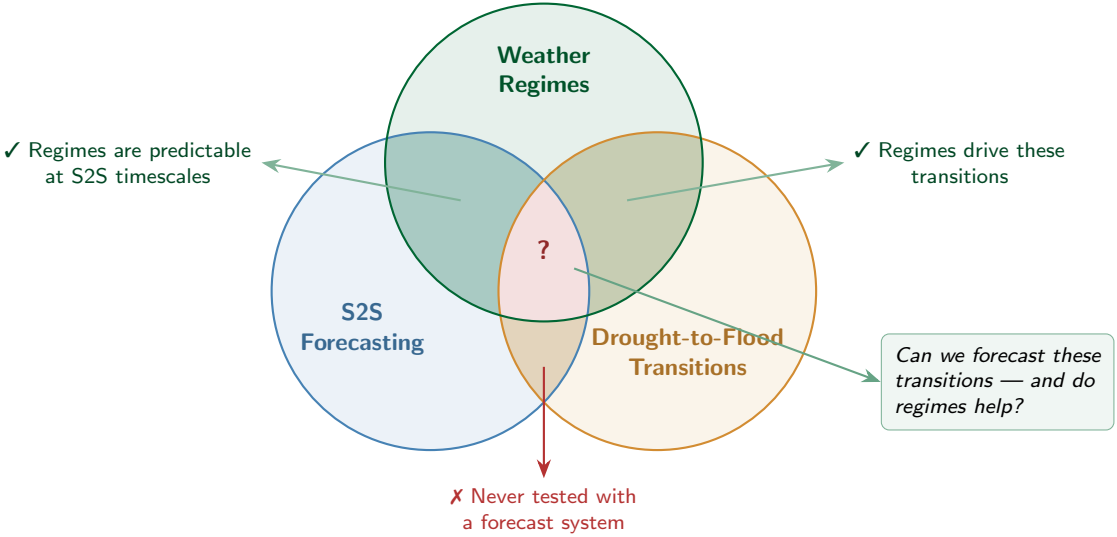
Transition

Rain hits dry soil
Runoff amplified

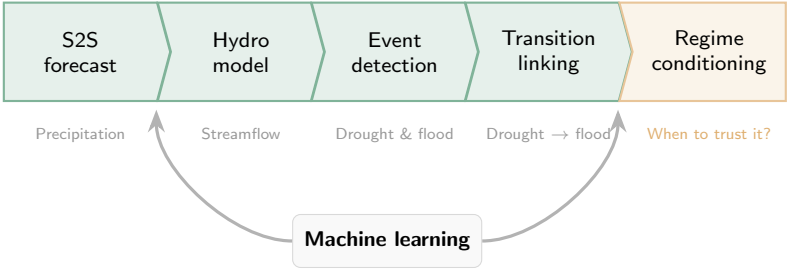


Adapted from DeFlorio et al. (2024)

Where Is the Gap?



How Do I Test This?



Blocking breaking down weeks 3–4, above-normal transition risk weeks 4–6

Which Forecast System?

Subseasonal (EEFH)

Newer physics
183 inits / year
11 members

↑ **Temporal sampling**

vs

Seasonal (SEAS5)

Older physics
12 inits / year
25 members

↑ **Ensemble sampling**

Ireland | 2006–2016 | t2m, tp, ws10

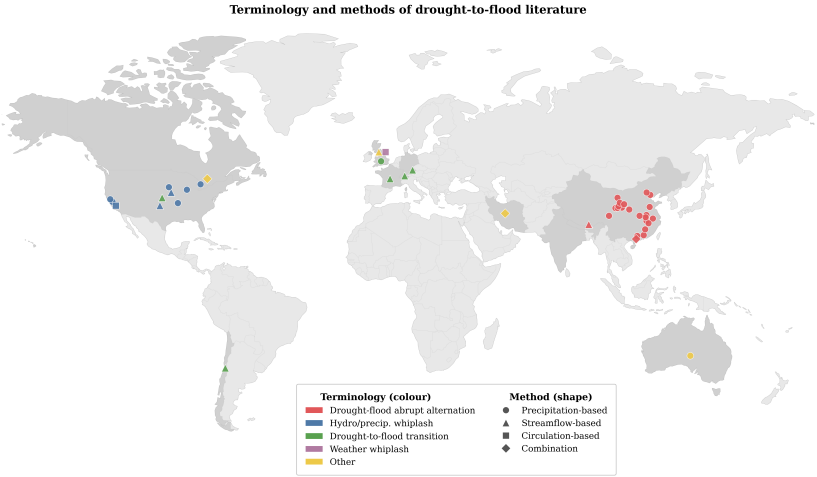
Verified against

Met Éireann station observations

ERA5-Land reanalysis

Deterministic scores: RMSE, bias, RMSS
Station and spatial verification

How Do We Detect These Events?



Takeaways

1. Drought-to-flood transitions are common, damaging, and unfold over the S2S timescale

2. The atmospheric patterns that cause them are the same ones that give S2S forecasts their skill

3. Nobody has tested whether you can use those regimes to predict drought-to-flood transitions

References

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- Yang, Y., G. Villarini, and L. Yang, 2025: Spatiotemporal dynamics of hydrological whiplash events across the contiguous United States. *ESS Open Archive* (preprint).